

Transformer-Enhanced SAC for Joint Power–Frequency Allocation in HTS Systems

Abstract—In future space–air–ground integrated networks (SAGIN), multi-beam high-throughput satellite (HTS) systems require efficient joint allocation of power and frequency resources under time-varying scenario to accommodate growing demand and enhance overall system capacity. To address the challenges of high-dimensional dynamic resource management, this paper proposes a Transformer-enhanced Soft Actor–Critic (T-SAC) algorithm. The resource allocation problem is formulated as a Satellite-Centric Markov Decision Process (SC-MDP), and a multi-head self-attention Transformer encoder is incorporated into the SAC framework as a global feature extractor to effectively capture cross-beam spatio-temporal dependencies and alleviate severe co-channel interference. The proposed method achieves superior higher sample efficiency, training stability, and stronger generalization in high-dimensional mixed action spaces. Simulation results show that T-SAC achieves significant improvements over baseline algorithms in average signal-to-interference-plus-noise ratio (SINR), satisfaction-gap-based fairness (SGM), and overall user satisfaction, demonstrating its robustness and efficiency in dynamic multi-beam environments.

Index Terms—High-throughput satellite, dynamic resource allocation, deep reinforcement learning, Soft Actor–Critic, transformer

I. INTRODUCTION

Multi-beam high-throughput satellite (HTS) systems have become a key enabler of future space–air–ground integrated networks (SAGIN). Through frequency reuse and flexible payload architectures, HTS systems can significantly enhance spectral efficiency and overall system capacity [1]–[3]. However, as the number of beams and user demand continue to grow, the joint allocation of power and frequency under time-varying scenarios has evolved into a complex NP-hard problem [4]. Geographically heterogeneous traffic distributions and severe co-channel interference further increase this complexity and jointly determine system throughput and user satisfaction. Consequently, intelligent, scalable, and adaptive resource allocation has become a central research challenge for next-generation satellite communication systems.

Early research primarily relied on convex optimization and mathematical programming approaches. Typical works formulated convex optimization models and employed sequential convex approximation (SCA) to maximize system throughput or minimize total power consumption under signal-to-interference-plus-noise ratio (SINR) and bandwidth constraints [3]. Although these methods can yield theoretically optimal or near-optimal solutions in small-scale geostationary orbit (GEO) systems, they depend heavily on accurate channel information and centralized computation. As the number of beams increases, their computational complexity grows

rapidly, limiting scalability and making them unsuitable for real-time HTS operations [2]. To alleviate the burden imposed by large beam counts, meta-heuristic algorithms—such as genetic algorithms (GA) [5], simulated annealing (SA) [6], and particle swarm optimization (PSO) [7]—have been introduced for joint resource allocation. However, these methods often suffer from slow convergence, sensitivity to initialization, and limited adaptability to time-varying environments.

In recent years, deep reinforcement learning (DRL) has emerged as a promising tool for tackling complex, dynamic satellite environments. Unlike model-driven methods, DRL does not require an explicit system model; instead, it learns resource allocation policies through interactions with the environment [1], [6], [8]–[11], and has demonstrated performance superior to convex optimization and heuristic algorithm in dynamic multi-beam scenarios. However, when these classical DRL algorithms are applied directly to HTS systems with hundreds of beams, the mixed power–frequency action space grows exponentially. This results in severely reduced exploration efficiency, sparse rewards, and unstable training, causing slow convergence or entrapment in suboptimal solutions. To enhance scalability, multi-agent DRL (MADRL) frameworks have been introduced to perform distributed joint decision-making [2], [12], [13]. Nonetheless, under constrained on-board computation and non-stationary multi-agent interactions, MADRL still suffers from heavy communication and training overhead, and limited convergence stability. Against the above limitations of traditional DRL methods, the Soft Actor–Critic (SAC) algorithm [14] offers higher sample efficiency and superior training stability, making it naturally well-suited for constructing a high-dimensional mixed-action resource allocation framework for multi-beam HTS systems.

On the other hand, existing DRL-based satellite resource allocation frameworks also face architectural limitations. Mainstream network architecture such as multilayer perceptron (MLP) and convolutional neural network (CNN) primarily extract local features and struggle to model global correlations among beams. As a result, their collaboration and generalization capability deteriorate significantly when traffic distribution or beam counts change. Transformer has recently emerged as a promising alternative [15]. By exploiting multi-head self-attention, Transformers can explicitly capture global and long-range spatio-temporal dependencies. As demonstrated in [16], Transformer-based models exhibit superior policy stability, exploration efficiency, and generalization in large action spaces compared with conventional deep neural networks.

This paper proposes a Transformer-enhanced Soft Actor–

Critic algorithm (T-SAC) for multi-beam HTS systems, designed for joint power-frequency resource allocation. The resource management problem is modeled as a Satellite-Centric Markov Decision Process (SCMDP). The proposed method builds upon the SAC framework to address power control and frequency planning in high-dimensional action spaces. A Transformer encoder with multi-head self-attention is integrated into the SAC architecture as a feature extraction module. Simulation results demonstrate that compared to baseline algorithms, the proposed T-SAC achieves significant performance gains in key metrics including average SINR, allocation fairness, and user satisfaction. This validates its effectiveness in achieving efficient and stable resource allocation within dynamic multi-beam HTS systems.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

Consider the downlink of a time-slotted, multi-beam, HTS communication system with a single GEO satellite. Let $\mathcal{T} = \{1, 2, \dots, T\}$ denote the set of time slot indices. The on-board beam-coverage schedule [17] is updated every Δt time units, so each time slot spans exactly one update interval. As shown in Fig.1, the multi-beam satellite can generate N_b beams, which are denoted as $\mathcal{B} = \{1, 2, \dots, N_b\}$ with the beamwidths of bw° . To reduce modeling complexity and highlight the core mechanism of the dynamic resource allocation strategy, we assume that within each cell determined by the beam-coverage schedule, the traffic demand of all users is represented by a superuser located at the beam-pointing center. Consequently, N_b cells are illuminated by these beams. Therefore, the traffic demand of the cell can be expressed as $\mathcal{D}(t) = \{td_1(t), td_2(t), \dots, td_{N_b}(t)\}$ in which $td_b(t)$ is the aggregated traffic demand of cell b at time t . Correspondingly, the actual traffic allocated by the satellite communication system is denoted as $\mathcal{A}(t) = \{tp_1(t), tp_2(t), \dots, tp_{N_b}(t)\}$ in which $tp_b(t)$ denotes the actual traffic allocated to cell b at time t .

The allocation and management of satellite resources is essential to enhance satellite system throughput and reduce

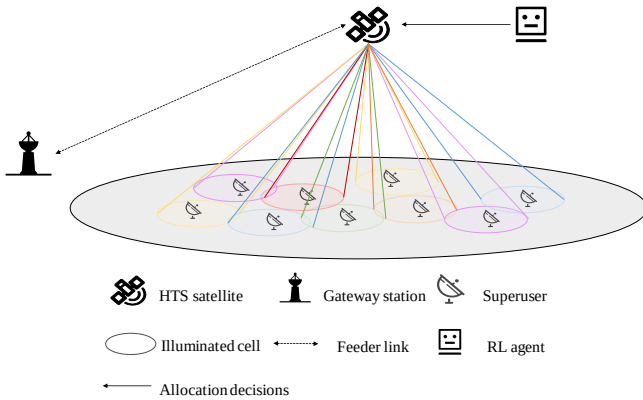


Fig. 1: System Model

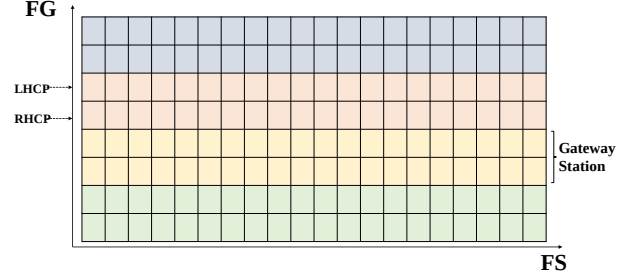


Fig. 2: Frequency resource pool

inter-beam co-channel interference. The satellite resource pool (SRP) is a centralized resource management framework on multi-beam satellite, which can be further subdivided into frequency resource pool (FRP) as shown in Fig.2 and power resource pool (PRP). The reuse mechanism of FRP and the concept of frequency group (FG) and frequency slot (FS) have been introduced in [9]. The multi-beam satellite operates in the Ka-band, spanning the downlink frequency range from f_{start} to f_{end} , which provides a total multiplexing bandwidth of $B_{\text{tot}} = f_{\text{end}} - f_{\text{start}}$. The FRP contains N_g FGs, denoted as $\mathcal{G} = \{1, 2, \dots, N_g\}$, where N_g is determined by the number of gateway stations. Odd- and even-indexed FGs belong to two mutually orthogonal polarization modes, defined as $polarization \in \{LHCP, RHCP\}$. Each FG is equally divided into N_s FSs, whose set can be presented as $\mathcal{S} = \{1, 2, \dots, N_s\}$, where the bandwidth of each FS is $B_{\text{slot}} = \frac{B_{\text{tot}}}{N_s}$. Each beam can only transmit one carrier, and the occupied bandwidth must be continuous and no more than B_{max} to prevent any single beam from monopolising excessive spectrum and keep the RF front-end within its practical linear operating range. The PRP distributes a suitable output power to each beam while respecting to limits. 1) A total power budget P_{tot} imposed by the satellite's overall energy balance. 2) A per-beam ceiling P_{max} , determined by the saturation rating of each traveling-wave tube amplifier (TWTA).

Considering that beam coverage scheme periodically changes the service beam, the satellite downlink will be modeled according to channel time-varying characteristics. $\forall t \in \mathcal{T}, \forall b \in \mathcal{B}, \forall s \in \mathcal{S}$, the total path loss between satellite and cell b in FS s is $L_b^{s,f}(t) = L_b^{s,f}(t)L_b^{s,r}(t)L_b^{s,c}(t)$ with free space path loss

$$L_b^{s,f}(t) = \left(\frac{4\pi d_b(t) f^s}{c} \right)^{-2}, \quad (1)$$

where $d_b(t)$ is the distance between the satellite and cell b at time t , f^s is the center frequency of FS s , and c is the speed of light. Rain and cloud attenuation, $L_b^{s,r}(t)$ and $L_b^{s,c}(t)$ are evaluated using [18] and [19], respectively. The satellite transmit antenna gain $G_b^{Tx}(\theta_{b,i}(t))$ is based on [20] with a boresight gain of G_b^{Tx} , where $\theta_{b,i}(t)$ is the off-axis angle between the boresight of beam b and the line-of-sight direction to cell i at time t . Because the main axis of cell b 's receiving

antenna is adapted to coincide with the satellite–user line-of-sight, cell b experiences the antenna’s boresight gain, denoted by G_b^{Rx} . The signal power transmitted from beam b to cell i on FS s at time t is:

$$\delta_{b,i}^s(t) = P_b^{Tx}(t) G_b^{Tx}(\theta_{b,i}(t)) G_i^{Rx} L_i^s(t) \quad (2)$$

where $P_b^{Tx}(t)$ is transmit power of beam b allocated by PRP at time t .

Then the signal-to-interference-plus-noise ratio (SINR) of FS s occupied by beam b pointing to cell b at time t can be expressed as:

$$\Gamma_b^s(t) = \frac{\delta_{b,b}^s(t)}{\sum_{i=1, i \neq b}^{N_b} x_{i,b}^s \delta_{i,b}^s(t) + k_B T_b B_{\text{slot}}}, \quad (3)$$

where $x_{i,j}^s$ is a binary indicator, $x_{i,j}^s = 1$ when the following three conditions are met simultaneously. 1) Beam i occupies FS s . 2) Beam i and j have the same polarization mode. 3) Beam i and j are in different FGs. Otherwise, $x_{i,j}^s = 0$. k_B denotes the Boltzmann constant. T_b refers to the receiving noise temperature of cell b . Thus, the traffic of cell b provided by beam b at time t is:

$$tp_b(t) = \sum_{k=1}^{N_s} B_{\text{slot}} \log_2(1 + \Gamma_b^k) \quad (4)$$

B. Problem Formulation

The main goal of this paper is to maximize the service satisfaction of all cells by matching the allocated traffic to their demand traffic as closely as possible. To quantitatively evaluate the effectiveness and fairness of the resource allocation strategy, the satisfaction-gap measure (SGM) proposed in [13] is adopted as an important performance metric, which can be expressed as:

$$\Lambda_b(t) = 1 - \left(1 - e^{-|\lambda_b(t)|}\right)^3 \quad (5)$$

where

$$\lambda_b(t) = \begin{cases} (\chi_b(t) - 1) + j \frac{\Delta_b(t)}{\beta}, & \chi_b(t) \geq 1 \\ (1 - \frac{1}{\chi_b(t)}) + j \frac{\Delta_b(t)}{\beta}, & 0 < \chi_b(t) < 1 \end{cases} \quad (6)$$

Specifically, the composite metric composed of the satisfaction index $\chi_b(t) = \frac{tp_b(t)}{td_b(t)}$ and the gap $\Delta_b(t) = tp_b(t) - td_b(t)$ imposes a heavier penalty on under-served allocation, prompting the algorithm to prioritize filling service deficits, while applying a milder penalty to over-served users to curb resource wastage.

Define $\mathbf{X} = [x_b^s(t)]_{T \times N_b \times N_s}$, $\mathbf{Y} = [y_b^g(t)]_{T \times N_b \times N_g}$, where $x_b^s(t) = 1$ when beam b occupies FS s , otherwise $x_b^s(t) = 0$, $y_b^g(t) = 1$, when beam b belongs to FG g , otherwise $y_b^g(t) = 0$, and $\mathbf{P} = [P_b^{Tx}(t)]_{T \times B}$. Therefore, the optimization problem is formulated as:

$$\mathbf{P1}: \max_{\mathbf{X}, \mathbf{Y}, \mathbf{P}} \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}} \epsilon_1 \Lambda_b(t) + \epsilon_2 \chi_b(t) \quad (7)$$

$$\text{s.t.} \sum_{b \in \mathcal{B}} y_b^g(t) x_b^s(t) \leq 1, \forall t \in \mathcal{T}, \forall g \in \mathcal{G}, \forall s \in \mathcal{S} \quad (7a)$$

$$\sum_{g \in \mathcal{G}} y_b^g(t) = 1, \forall t \in \mathcal{T}, \forall b \in \mathcal{B}, \forall s \in \mathcal{S} \quad (7b)$$

$$0 < \sum_{s=1}^{N_s-1} |x_b^{s+1}(t) - x_b^s(t)| \leq 2, \forall t \in \mathcal{T}, \forall b \in \mathcal{B} \quad (7c)$$

$$\sum_{s \in \mathcal{S}} x_b^s(t) \cdot B_{\text{slot}} \leq B_{\text{max}}, \forall t \in \mathcal{T}, \forall b \in \mathcal{B} \quad (7d)$$

$$P_b^{Tx}(t) \leq P_{\text{max}}, \forall t \in \mathcal{T}, \forall b \in \mathcal{B} \quad (7e)$$

$$\sum_{b \in \mathcal{B}} P_b^{Tx}(t) \leq P_{\text{tot}} \quad (7f)$$

where ϵ_1 and ϵ_2 are tunable factors that can be adjusted to accommodate different service scenarios. (7a) restricts each FS to be occupied by at most one single beam. (7b) restricts each beam can occupy one single FG. (7c) ensures continuity in frequency allocation. (7d) limits the bandwidth allocated to each beam to no more than B_{max} . (7e) limits the transmit power allocated to each beam to at most P_{max} . (7f) guarantees that the total allocated does not exceed overall power budget P_{tot} .

The problem P1 is a large-scale, time-coupled and non-convex optimization problem that is challenging for the classical deterministic optimization algorithm. To cope with this complexity, P1 can be further modeled as a satellite-centric Markov decision process (SC-MDP) and solved by proposed Transformer-enhanced Soft Actor-Critic (T-SAC) algorithm.

III. DYNAMIC RESOURCE ALLOCATION BASED ON T-SAC ALGORITHM

The RL agent is first trained at the ground gateway until convergence and then deployed on the satellite, where it operates in real time to make resource allocation decisions. Updated models can be transmitted from the gateway periodically as new training data arrive.

A. SC-MDP

Resource allocation decisions are modeled as a Markov decision process (MDP): each time slot is regarded as an episode containing N_b timesteps, at every timestep the agent allocates resource to the beam that is currently illuminated.

1) *State Space*: The satellite communication constitutes the RL environment; therefore, the agent is assumed to have full observation of the global state required to make decisions. The beam state consists of the beam traffic demand $td_b(t)$, the beam position $\Phi_b(t)$. The PRP state includes the remaining available power $\xi_b(t) = P_{\text{tot}} - \sum_{i=1}^{b-1} P_i^{Tx}(t)$. The FRP state includes the FS occupancy status $\mathcal{O}_b(t) = \mathbf{Y}_b^T(t) \cdot \mathbf{X}_b(t)$. The channel state includes the interfere power of each FS $\mathcal{I}_b(t) = \{\zeta_b^1(t), \zeta_b^2(t), \dots, \zeta_b^{N_s}(t)\}$, where $\zeta_b^s(t) = \sum_{i=1}^{b-1} x_i^s \delta_{i,b}^s(t)$. Then the state space can be formulated as:

$$\mathcal{S}_b(t) = (td_b(t), \Phi_b(t), \xi_b(t), \mathcal{O}_b(t), \mathcal{I}_b(t)). \quad (8)$$

2) *Action Space*: The action space $\mathcal{A}$ includes frequency assignment and power allocation. To improve training stability, two tricks aiming to reduce the dimensionality are applied to the proposed algorithm: 1) FGs are assigned cyclically, that is $y_b^{(b \bmod N_g)}(t) = 1$, which allows beam to be evenly distributed across different FGs, and because the FG index is statically set by the environment. The agent does not need to make decisions along the dimension, reducing the action space by a factor of N_g . 2) Continuous power is quantized evenly into ω discrete levels, thereby avoiding the unnecessary complexity of a hybrid continuous-discrete action space. The action space can be formulated as:

$$\mathcal{A}_b(t) = (\gamma_b(t), \kappa_b(t), P_b^{Tx}(t)), \quad (9)$$

where $n_{\max} = \frac{B_{\max}}{B_{\text{slot}}}$ is the maximum number of FSs available per single beam, $\gamma_b(t) \in \{1, 2, \dots, N_s\}$ is the assigned index of the starting FS, $\kappa_b(t) \in \{1, 2, \dots, n_{\max}\}$ is the assigned FSs number, $P_b^{Tx}(t) \in \{\frac{P_{\max}}{\omega}, \frac{2 \cdot P_{\max}}{\omega}, \dots, P_{\max}\}$. The action space shrinks from $N_g \times N_s \times n_{\max} \times \omega$ to $N_s \times n_{\max} \times \omega$.

3) *Reward*: According to problem P1, the reward is defined as:

$$r_b(t) = \Lambda_b(t) + \chi_b(t) + \Psi_b(t), \quad (10)$$

where $\Psi_b(t)$ is the penalty, defined as:

$$\Psi_b(t) = \begin{cases} -R_1, & (7a) \text{ is violated} \\ -R_2, & (7f) \text{ is violated,} \end{cases} \quad (11)$$

R_1 and R_2 are two constants.

B. The Design of T-SAC

The overall architecture of the T-SAC algorithm is shown in Fig.3. In the T-SAC architecture, a Transformer encoder in [15] serves as the feature extractor. The raw state is semantically decomposed into several sub-features: scalar attributes $\{td_b(t), \Phi_b(t), \xi_b(t)\}$ and vector attributes $\{\mathcal{O}_b(t), \mathcal{I}_b(t)\}$. Each sub-feature is projected to a common hidden size d_{model}

by separate linear layers and treated as an independent token, while sequential tokens $[td_b(t), \Phi_b(t), \xi_b(t), \mathcal{O}_b(t), \mathcal{I}_b(t)]$ receive learnable positional embeddings. The resulting token sequence is fed into the Transformer encoder, where multi-head self-attention jointly models scalar-vector and inter-vector interactions, capturing long-range cross-band interference patterns and fusing global power constraints with local beam details. It can be described as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \quad (12)$$

where $\mathbf{Q}$, $\mathbf{K}$ and $\mathbf{V}$ represent the query, key and value, respectively, all derived from the same input token embeddings by three independent learnable projections $W_{\mathbf{Q}}$, $W_{\mathbf{K}}$ and $W_{\mathbf{V}}$. Shared encoder parameters allow the model to adapt to different beam number and interference levels without structural changes, endowing the algorithm with strong generalization capability.

SAC proposed in [14] is served as the decision-making core, using the maximum-entropy principle to learn an optimal policy over the discrete action space $\mathcal{A}$. By augmenting the reward with the policy entropy

$$\mathcal{H}(\pi) = \mathbb{E}_{a \sim \pi}[-\log \pi(a | s)], \quad (13)$$

Algorithm 1 Transformer-enhanced Soft Actor-Critic (T-SAC) Training

Input: Replay-buffer capacity C , learning rates $(\eta_\theta, \eta_{\psi_1}, \eta_{\psi_2})$, discount γ , target-update rate τ , target entropy $\mathcal{H}$

Output: Trained actor π_θ and two critics Q_{ψ_1}, Q_{ψ_2}

- 1: Initialize actor π_θ , critics Q_{ψ_1}, Q_{ψ_2} and target critics $\bar{Q}_{\psi_1} \leftarrow Q_{\psi_1}$
 - 2: Initialize log-temperature parameter $\log \alpha_{\text{init}}$
 - 3: Initialize empty replay buffer $\mathcal{D}$ with capacity C
 - 4: **for** $t = 1$ **to** T **do**
 - 5: Reset environment and obtain initial state $s_1(t)$
 - 6: **for** beam index $b = 1$ **to** N_b **do**
 - 7: Execute $a_b(t)$, get reward $r_b(t)$, next state $s_{b+1}(t)$ and flag *done*
 - 8: Store transition $(s_b(t), a_b(t), r_b(t), s_{b+1}(t), \text{done})$ in $\mathcal{D}$
 - 9: **if** $|\mathcal{D}| > \text{min_size}$ **then**
 - 10: Sample a mini-batch $\mathbb{B}$ from $\mathcal{D}$
 - 11: Compute TD target $\mu_b(t)$ with Eq.(18)
 - 12: Update critics by minimizing Eq. (15) using η_{ψ_i}
 - 13: Update actor by minimizing Eq. (16) using η_θ
 - 14: Update temperature α via Eq. (17) using η_α
 - 15: Soft-update targets: $\bar{Q}_{\psi_i} \leftarrow (1 - \tau)\bar{Q}_{\psi_i} + \tau Q_{\psi_i}$
 - 16: **end if**
 - 17: **if** *done* **then**
 - 18: **break**
 - 19: **end if**
 - 20: **end for**
 - 21: **end for**
 - 22: **return** $\pi_\theta, Q_{\psi_1}, Q_{\psi_2}$
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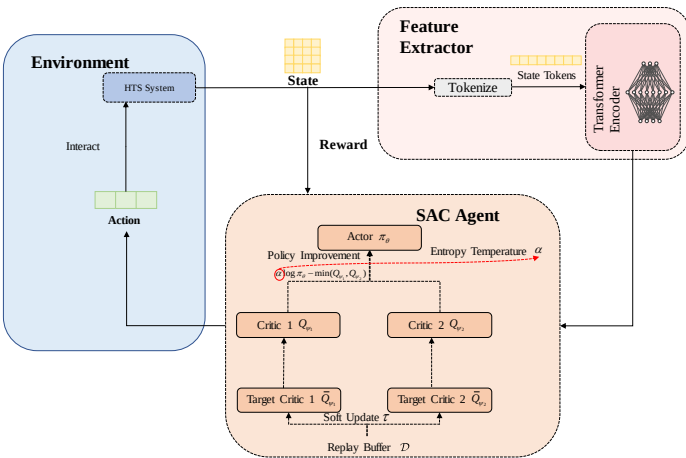


Fig. 3: The overall architecture of T-SAC

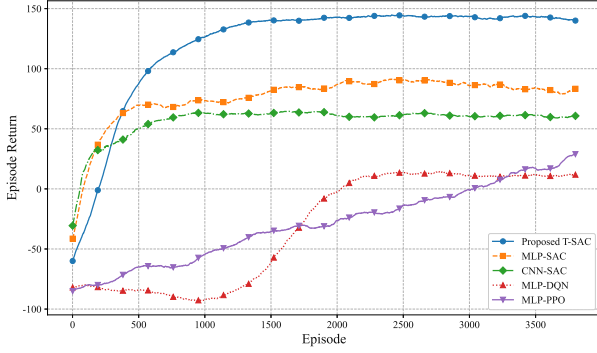


Fig. 4: Training convergence curves of various algorithms

SAC explicitly encourages policy stochasticity via a maximum-entropy objective. In large action spaces it explores a more diverse set of actions than conventional methods, thereby reducing the risk of prematurely converging to sub-optimal solutions. The objective function is

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{b=1}^{N_b} \gamma^{b-1} (r_b(t) + \alpha \mathcal{H}(\pi(\cdot | s_b(t)))) \right], \quad (14)$$

where γ is the discount factor, α is a learnable temperature used to make the actual entropy approach the target entropy. SAC consists of three learnable core components: the actor $\pi_\theta(a_{t,b} | s_{t,b})$, the critic Q_{ψ_1} , Q_{ψ_2} which are designed to suppress overestimation, and the temperature parameter α . The update process of parameters ψ_i , θ and α are

$$\psi_i \leftarrow \arg \min_{\psi_i} (Q_{\psi_i}(s_b(t), a_b(t)) - \mu_b(t))^2, i \in \{1, 2\}, \quad (15)$$

$$\theta \leftarrow \arg \min_{\theta} [\alpha \log \pi_\theta(a_b(t) | s_b(t)) - \min_i Q_{\psi_i}(s_b(t), a_b(t))], \quad (16)$$

$$\alpha \leftarrow \arg \min_{\alpha} [-\alpha (\log \pi_\theta(a_b(t) | s_b(t)) + \bar{\mathcal{H}})], \quad (17)$$

where

$$\mu_b(t) = r_b(t) + \gamma (\min_i \bar{Q}_{\psi_i}(s_b(t), a'_b(t)) - \alpha \log \pi_\theta(a_b(t) | s_b(t))), \quad (18)$$

$$a'_{b+1}(t) \sim \pi_\theta(\cdot | s_{b+1}(t)). \quad (19)$$

IV. SIMULATION RESULTS AND DISCUSSION

In this section, several simulation experiments are carried out to verify the effectiveness of the proposed algorithm. First, a multi-beam High Throughput Satellite (HTS) communication system simulation environment is established using Python, and a reinforcement learning framework is implemented in PyTorch, where an agent interacts with the environment to generate joint power–frequency allocation strategies. In the simulation setup, the HTS beams cover 100 major cities across China. The traffic demand for each beam is randomly generated, and the beams are indexed in descending order according to their traffic demands to simulate the uneven load distribution across regions. The main simulation parameters and the main T-SAC parameters are summarized in Table 1 and Table 2, respectively.

To ensure the fairness and comprehensiveness of the evaluation, this paper implements seven typical baseline algorithms under the same simulation conditions, as follows:

1) **Fixed Scheme**: Each beam is assigned a fixed number of FS and a constant transmit power, with the starting FS and FG selected sequentially.

2) **Greedy Scheme**: In each allocation, the algorithm selects the FS with the least occupied beam based on the current FS occupancy status. The transmit power is adaptively selected from three levels $P_b^{Tx}(t) \in \{20W, 25W, 30W\}$ based on traffic demand.

3) **Random Scheme**: By uniformly sampling frequency and power combinations in the legal action space, serving as a reference for the lower limit of performance.

4) **MLP-SAC**: SAC based on multilayer perception (MLP).

5) **CNN-SAC**: SAC based on convolutional neural networks (CNN).

6) **MLP-DQN**: Deep Q-Network (DQN) based on MLP. It employs a factorized three-head design to separately estimate Q-values for the discrete action spaces of frequency, time-slot, and power. Each branch independently predicts the action-value function of its respective dimension, and the overall joint Q-value is obtained as the sum of the three components.

7) **MLP-PPO**: Proximal Policy Optimization (PPO) based on MLP. MLP-PPO adopts an Actor–Critic architecture. An Actor network outputs three independent discrete action distributions corresponding to frequency, time-slot, and power, while a Critic network estimates the state value.

Fig.5 shows the mean and variance of the SINR for each algorithm in 10 random scenarios. The proposed T-SAC achieves the highest average SINR and the smallest variance in all scenarios, indicating its optimal performance in interference

TABLE I: Main Simulation Parameters

System Parameters	
Frequency Band	Ka-band 17.7-20.2 GHz
Number of FGs N_g	8
Number of FSs N_s	100
GEO Satellite	
Orbit Longitude	122.2° East
Orbit Altitude	35786 km
Number of Beams N_b	100
Beamwidth	0.5°
Total Power Budget P_{tot}	3000 W
Transmitting Antenna Gain G^{Tx}	50 dBi
Number of Power levels	10
User Equipment	
Receiving Antenna Gain G^{Rx}	40 dBi
Noise Temperature T	290 K

TABLE II: Main T-SAC Parameters

Parameter	Value
Training Episodes	4000
Minimum Sampling Steps	2000 steps
Replay-Buffer Capacity C	20000 steps
Batch Size $\mathbb{B}$	256
Discount Factor γ	0.99
Target-update Rate τ	0.005
Target Entropy $\bar{\mathcal{H}}$	-6.9
Learning Rate $[\eta_\theta, \eta_{\psi_i}, \eta_\alpha]$	$[10^{-5}, 10^{-4}, 10^{-4}]$

suppression and link stability. MLP-SAC and greedy scheme have average SINR of approximately 14–15 dB, significantly lower than T-SAC. Fixed scheme has a slightly lower average SINR of approximately 12–14 dB due to the lack of flexibility. CNN-SAC has an average SINR of approximately 12 dB and a large variance, indicating that CNN struggles to capture the intrinsic dependence of the system and dynamic features of the beam. DQN suffers from low learning efficiency in high-dimensional spaces due to its limited exploration capabilities, resulting in poor performance in interference suppression. PPO and the Random scheme have similar performance and are the worst, indicating that PPO struggles to effectively explore and learn optimal strategies in high-dimensional action space scenarios, failing to demonstrate the learning capabilities of reinforcement learning algorithms and only approaching the level of random exploration. Overall, T-SAC improves the average CINR by about 2–3 dB and reduces the variance by more than 70% compared to the optimal baseline (MLP-SAC), validating its global coordination capability.

Fig.6 shows T-SAC, in particular, exhibits an average SGM exceeding 0.9 and a variance below 0.05, demonstrating optimal resource utilization efficiency and strong robustness. The average SGM of MLP-SAC is approximately 0.7–0.8, while the average SGM of CNN-SAC, MLP-DQN, and the Greedy scheme is approximately 0.6–0.7. The fixed and random schemes have an average SGM of only about 0.5. Experimental results demonstrate that learning algorithms can learn the intrinsic mechanisms for improving resource allocation fairness through policy optimization, thus significantly outperforming non-learning algorithms in this dimension. Particularly, this indicates that T-SAC, through a self-attention mechanism, achieves information modeling based on global context, thereby realizing more efficient and fairer allocation of spectrum resources and power.

Fig.7 shows the T-SAC algorithm maintains a high satisfaction score close to 1.0 across all scenarios, with a variance below 0.05, indicating its ability to stably meet the service requirements of different beams under dynamic environment, demonstrating excellent robustness and generalization ability. The average satisfaction scores of MLP-SAC, greedy and fixed scheme range from 0.85 to 0.9, while the average satisfaction score of the CNN-SAC is approximately 0.8, which can meet user needs to a certain extent. However, due to the lack of global dependency modeling, it is difficult to guarantee allocation fairness. The satisfaction scores of DQN, PPO, and random scheme are all significantly low, reflecting their difficulty in achieving effective resource coordination in complex dynamic environments.

V. CONCLUSION

This paper proposes a T-SAC algorithm for multi-beam HTS systems to achieve joint power-frequency resource allocation. By modeling the dynamic satellite resource allocation problem as an SC-MDP, T-SAC introduces a Transformer encoder to capture long-range cross-band interference patterns and fusing global power constraints with local beam details. Simulation

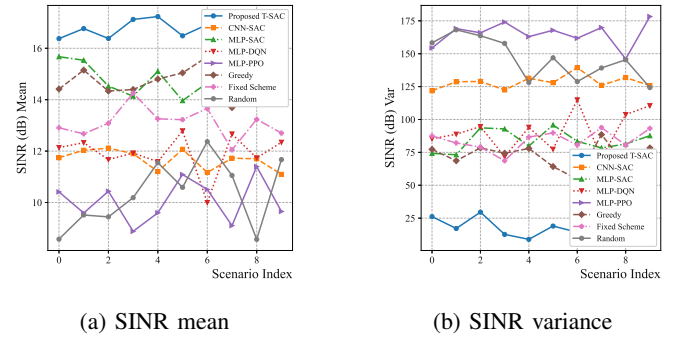


Fig. 5: Comparison of SINR mean and variance of various algorithms

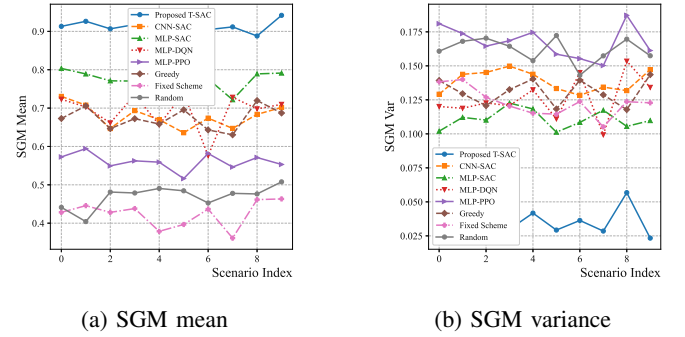


Fig. 6: Comparison of SGM mean and variance of various algorithms

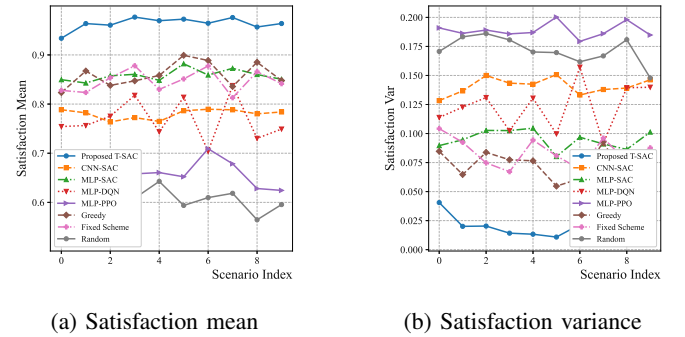


Fig. 7: Comparison of satisfaction mean and variance of various algorithms

results show that T-SAC significantly outperforms traditional heuristic methods and existing reinforcement learning baselines under various stochastic scenarios. The algorithm exhibits strong robustness in both resource utilization efficiency and interference suppression. In summary, the proposed T-SAC framework, through self-attention representation learning, effectively uncovers the intrinsic mechanisms of global spectrum coordination and dynamic interference management, providing a feasible and efficient solution for intelligent satellite resource scheduling in next-generation space-ground integrated networks.

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